

Stability Analysis of Interval Type-2 Fuzzy-Model-Based Control Systems

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Abstract—This paper presents the stability analysis of interval type-2 fuzzy-model-based (FMB) control systems. To investigate the system stability, an interval type-2 Takagi–Sugeno (T-S) fuzzy model, which can be regarded as a collection of a number of type-1 T-S fuzzy models, is proposed to represent the nonlinear plant subject to parameter uncertainties. With the lower and upper membership functions, the parameter uncertainties can be effectively captured. Based on the interval type-2 T-S fuzzy model, an interval type-2 fuzzy controller is proposed to close the feedback loop. To facilitate the stability analysis, the information of the footprint of uncertainty is used to develop some membership function conditions, which allow the introduction of slack matrices to handle the parameter uncertainties in the stability analysis. Stability conditions in terms of linear matrix inequalities are derived using a Lyapunov-based approach. Simulation examples are given to illustrate the effectiveness of the proposed interval type-2 FMB control approach.

Index Terms—Fuzzy control, interval type-2 fuzzy model, stability analysis.

I. INTRODUCTION

TYPE-1 fuzzy-model-based (FMB) control systems [1] have drawn the attention of fuzzy control researchers for many years. The type-1 Takagi–Sugeno (T-S) fuzzy model [2], [3] offers a general framework for system analysis and controller synthesis. A fuzzy controller [4] was proposed to close the feedback loop. Basic stability conditions [4] in terms of linear matrix inequalities (LMIs) were derived based on the Lyapunov stability theory to guarantee system stability. The importance of the membership function of the type-1 T-S fuzzy model to system stability was revealed in [4]–[6]. By sharing the same membership functions between a type-1 fuzzy model and a fuzzy controller, relaxed stability conditions were achieved. Further relaxed stability conditions were accomplished and reported in [7]–[10]. It has been reported that the stability conditions in [7]–[10] are particular cases of [10] of which the stability conditions derived based on Polya’s theorem provided the least conservative result.

As the membership functions for the type-1 fuzzy sets contain no uncertainty information, the control problem cannot be directly handled if the nonlinear plant is subject to parameter

uncertainties. Moreover, the type-1 T-S fuzzy models employed in [4]–[10] are developed using type-1 fuzzy sets such that the grades of membership are certain. Once parameter uncertainties of nonlinear plants are considered, the grades of membership may become uncertain in value. Consequently, the nice properties leading to relaxed stability conditions will vanish, and Lyapunov-based stability analysis will lead to conservative stability conditions. In [11]–[13], a type-2 fuzzy logic system (FLS) was proposed to deal with uncertain grades of membership. The type-2 FLS can be regarded as a collection of theoretically infinite number of type-1 FLSs. As a result, additional information including system parameter uncertainty can be captured by the type-2 FLS [11]–[13]. Some theories of interval type-2 fuzzy logic [11]–[13] have been developed to theoretically support the design and implementation of type-2 FLSs. The superiority of interval type-2 fuzzy sets over type-1 fuzzy sets on dealing with uncertain grades of membership for various applications has been shown in [14]–[16]. The applications include adaptive filtering [14], autonomous mobile robots [15], and dc–dc power converters [16].

Despite their capability and nice properties, the type-2 fuzzy sets, which are still an emerging area of vigorous research activities, are more complex and computationally intensive [11]–[13]. In particular, the outstanding feature of the type-2 FLS on handling uncertainty offers a potential vehicle for carrying out the stability analysis of FMB control systems subject to parameter uncertainties. The stability analysis of interval type-2 FMB control systems subject to parameter uncertainties is an interesting research topic to explore. We are motivated to investigate the system stability of interval type-2 FMB control systems, which exhibits the attractive property for handling nonlinear plants subject to parameter uncertainties. It is desirable to obtain the stability analysis result under interval type-2 fuzzy sets and put the interval type-2 FMB control approach into practice.

In this paper, we generalized and extended our preliminary analysis result in [19] to interval type-2 FMB control systems. An interval type-2 fuzzy model is proposed to represent a nonlinear plant subject to parameter uncertainties. With the lower and upper membership functions, the parameter uncertainties can be effectively captured. The interval type-2 T-S fuzzy model can be regarded as a collection of a number of type-1 T-S fuzzy models. The information contained in the footprint of uncertainty (FOU) eliminates the favorable property of the type-1 FMB analysis approach to facilitate the stability analysis and consequently lead to a conservative stability analysis result. To facilitate the stability analysis, the

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lower and upper membership functions are employed to implement the proposed interval type-2 fuzzy controller. With such information, slack matrices can be introduced to alleviate the conservativeness of the stability analysis results. Based on the Lyapunov stability theory, stability conditions in terms of LMIs are derived to achieve a stable interval type-2 FMB control system.

This paper is organized as follows. In Section II, the interval type-2 T-S fuzzy model and the type-2 fuzzy controller are presented. In Section III, LMI-based stability conditions are derived. In Section IV, simulation examples are given to illustrate the merits of the proposed approach. Conclusions are drawn in Section V.

II. INTERVAL TYPE-2 T-S FUZZY MODEL AND FUZZY CONTROLLER

A nonlinear plant subject to parameter uncertainties is considered and represented by an interval type-2 T-S fuzzy model. An interval type-2 fuzzy controller is proposed to close the feedback loop to form an interval type-2 T-S FMB control system.

A. Interval Type-2 T-S Fuzzy Model

Consider an interval type-2 T-S fuzzy model [14] with p rules of which the antecedents are type-2 fuzzy sets, and the consequent is a crisp number (a linear dynamical system). The rule is of the following format:

Rule i : If $f_1(\mathbf{x}(t))$ is $\tilde{M}_1^i$ and, $\dots$, and $f_\Psi(\mathbf{x}(t))$ is $\tilde{M}_\Psi^i$
then $\dot{\mathbf{x}}(t) = \mathbf{A}_i \mathbf{x}(t) + \mathbf{B}_i \mathbf{u}(t)$ (1)

where $\tilde{M}_\alpha^i$ is an interval type-2 fuzzy set of rule i corresponding to the function $f_\alpha(\mathbf{x}(t))$, $\alpha = 1, 2, \dots, \Psi$; $i = 1, 2, \dots, p$; Ψ is a positive integer; $\mathbf{A}_i \in \mathbb{R}^{n \times n}$ and $\mathbf{B}_i \in \mathbb{R}^{n \times m}$ are known constant system and input matrices, respectively; $\mathbf{x}(t) \in \mathbb{R}^{n \times 1}$ is the system-state vector; and $\mathbf{u}(t) \in \mathbb{R}^{m \times 1}$ is the input vector. The firing strength of the i th rule is the following interval sets:

$$\tilde{w}_i(\mathbf{x}(t)) = [w_i^L(\mathbf{x}(t)), w_i^U(\mathbf{x}(t))], \quad i = 1, 2, \dots, p \quad (2)$$

where

$$w_i^L(\mathbf{x}(t)) = \mu_{\tilde{M}_1^i}(f_1(\mathbf{x}(t))) \times \mu_{\tilde{M}_2^i}(f_2(\mathbf{x}(t))) \times \dots \times \mu_{\tilde{M}_\Psi^i}(f_\Psi(\mathbf{x}(t))) \geq 0 \quad (3)$$

$$w_i^U(\mathbf{x}(t)) = \bar{\mu}_{\tilde{M}_1^i}(f_1(\mathbf{x}(t))) \times \bar{\mu}_{\tilde{M}_2^i}(f_2(\mathbf{x}(t))) \times \dots \times \bar{\mu}_{\tilde{M}_\Psi^i}(f_\Psi(\mathbf{x}(t))) \geq 0 \quad (4)$$

in which $\mu_{\tilde{M}_\alpha^i}(f_\alpha(\mathbf{x}(t))) \in [0, 1]$ and $\bar{\mu}_{\tilde{M}_\alpha^i}(f_\alpha(\mathbf{x}(t))) \in [0, 1]$ denote the lower and upper grades of membership governed by the lower and upper membership functions, respectively. Furthermore, it exhibits the property that $\bar{\mu}_{\tilde{M}_\alpha^i}(f_\alpha(\mathbf{x}(t))) \geq$

$\mu_{\tilde{M}_\alpha^i}(f_\alpha(\mathbf{x}(t)))$, which leads to $w_i^U(\mathbf{x}(t)) \geq w_i^L(\mathbf{x}(t))$ for all i . The inferred interval type-2 T-S fuzzy model is defined as

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \sum_{i=1}^p w_i^L(\mathbf{x}(t)) \underline{v}_i(\mathbf{x}(t)) (\mathbf{A}_i \mathbf{x}(t) + \mathbf{B}_i \mathbf{u}(t)) \\ &\quad + \sum_{i=1}^p w_i^U(\mathbf{x}(t)) \bar{v}_i(\mathbf{x}(t)) (\mathbf{A}_i \mathbf{x}(t) + \mathbf{B}_i \mathbf{u}(t)) \\ &= \sum_{i=1}^p w_i(\mathbf{x}(t)) (\mathbf{A}_i \mathbf{x}(t) + \mathbf{B}_i \mathbf{u}(t)) \end{aligned} \quad (5)$$

where

$$\begin{aligned} w_i(\mathbf{x}(t)) &= w_i^L(\mathbf{x}(t)) \underline{v}_i(\mathbf{x}(t)) \\ &\quad + w_i^U(\mathbf{x}(t)) \bar{v}_i(\mathbf{x}(t)) \in [0, 1] \\ \sum_{i=1}^p w_i(\mathbf{x}(t)) &= 1 \end{aligned} \quad (6)$$

in which $\underline{v}_i(\mathbf{x}(t)) \in [0, 1]$ and $\bar{v}_i(\mathbf{x}(t)) \in [0, 1]$ are nonlinear functions and exhibit the property of $\underline{v}_i(\mathbf{x}(t)) + \bar{v}_i(\mathbf{x}(t)) = 1$ for all i . In [14], the functions of $\underline{v}_i(\mathbf{x}(t))$ and $\bar{v}_i(\mathbf{x}(t))$ are both defined as 0.5. As the nonlinear plant is subject to parameter uncertainties, $\underline{v}_i(\mathbf{x}(t))$ and $\bar{v}_i(\mathbf{x}(t))$ may be dependent on the parameter uncertainties and not necessary to be known in this investigation. It should be noted that the interval type-2 T-S fuzzy model serves as a tool to aid the design of the type-2 fuzzy controller and is not necessarily implemented. Referring to the interval type-2 T-S fuzzy model of (5), the unknown functions of $\underline{v}_i(\mathbf{x}(t))$ and $\bar{v}_i(\mathbf{x}(t))$ will lead to uncertain $w_i(\mathbf{x}(t))$. Consequently, the stability analysis in [4]–[10], which requires $w_i(\mathbf{x}(t))$ to be known for all i , cannot be applied to investigate the system stability and design the fuzzy controller. In this paper, the lower and upper membership functions provide additional information to aid the design of the interval type-2 fuzzy controller and the conduction of the stability analysis.

A simple example is given below to illustrate the interval type-2 T-S fuzzy model. Consider the following simple scalar system subject to the uncertain parameter of $a(t)$:

$$\dot{x}(t) = \sin(a(x(t))x(t))x(t) \quad (7)$$

where $x(t) \in [-2, 2]$ is the system state. It is assumed that $a(x(t)) = (x(t)^2 + 1)/10$ has the known bounds of $\underline{a} \leq a(x(t)) \leq \bar{a}$, where $\underline{a} = 0.1$ and $\bar{a} = 0.5$ are the constant lower and upper bounds of $a(x(t))$, respectively. From (7), it can be seen that the lower and upper bounds for $\sin(a(x(t))x(t))$ are -1 and 1 , respectively. By following the way proposed for deriving the type-1 T-S fuzzy model [4] and denoting the membership functions as $\mu_{M_1^1}(x(t), a(x(t)))$ and $\mu_{M_2^1}(x(t), a(x(t))) = 1 - \mu_{M_1^1}(x(t), a(x(t)))$, we have $\mu_{M_1^1}(x(t), a(x(t)))(-1) + (1 - \mu_{M_1^1}(x(t), a(x(t))))(1) = \sin(a(x(t))x(t))$. Reshuffling the terms, we have $\mu_{M_1^1}(x(t), a(x(t))) = (1 - \sin(a(x(t))x(t)))/2$ and $\mu_{M_2^1}(x(t), a(x(t))) = (1 + \sin(a(x(t))x(t)))/2$. Considering

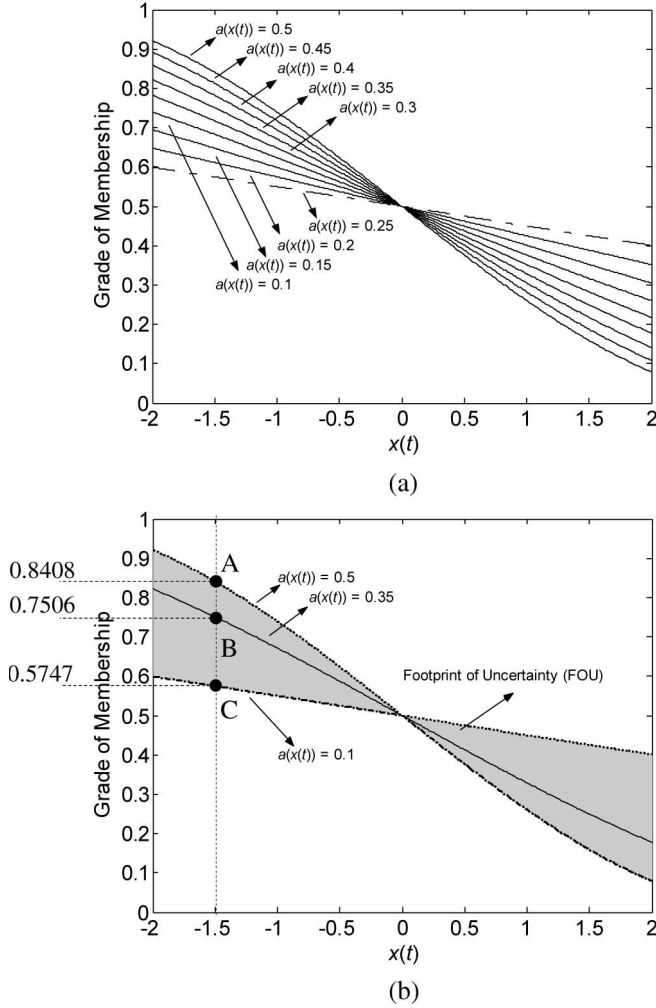


Fig. 1. Plot of $\mu_{M_1^1}(x(t), a(x(t))) = (1 - \sin(a(x(t))x(t)))/2$ with various values of $a(x(t))$, and illustration of FOU, lower and upper membership functions. (a) $\mu_{M_1^1}(x(t), a(x(t)))$. (b) Lower membership function $\underline{\mu}_{\tilde{M}_1}(x(t))$ (dotted line), upper membership function $\bar{\mu}_{\tilde{M}_1}(x(t))$ (dash-dot line), and membership function $\mu_{M_1^1}(x(t))$ with $a(t) = 0.35$ (solid line).

$a(x(t))$ to be a constant, we have the following type-1 fuzzy rule [4] to describe the system of (7):

$$\text{Rule } i: \text{ If } x(t) \text{ is } M_1^i, \text{ then } \dot{x}(t) = A_i x(t), \quad i = 1, 2 \quad (8)$$

where M_1^1 and M_1^2 are type-1 fuzzy sets; $A_1 = -1$ and $A_2 = 1$. The type-1 T-S fuzzy model [4] is defined as

$$\dot{x}(t) = \sum_{i=1}^2 w_1(x(t)) A_i x(t) \quad (9)$$

where the normalized grades of membership are defined as $w_i(x(t)) = \mu_{M_1^i}(x(t)) / (\mu_{M_1^1}(x(t)) + \mu_{M_1^2}(x(t)))$, $i = 1, 2$ [as $a(x(t))$ is assumed to be a constant, it is not a parameter of the type-1 membership functions]. Fig. 1(a) shows the type-1 membership functions of $\mu_{M_1^1}(x(t), a(x(t))) = (1 - \sin(a(x(t))x(t)))/2$ subject to different values of $a(x(t))$.

When $a(x(t))$ is considered as an uncertain value in the range of $\underline{a}$ and $\bar{a}$, it can be imagined that $\mu_{M_1^1}(x(t))$ is no longer a crisp membership function but characterized by the lower and

upper memberships of $\underline{\mu}_{\tilde{M}_1}(x(t))$ and $\bar{\mu}_{\tilde{M}_1}(x(t))$, respectively. Under such a situation, an interval type-2 T-S fuzzy model is employed to represent the nonlinear plant of (7) subject to parameter uncertainty of $a(x(t))$. With the information of the type-1 membership functions, the lower and upper membership functions can be obtained as follows. We have $\underline{\mu}_{\tilde{M}_1}(x(t)) \leq \mu_{M_1^1}(x(t), a(x(t))) \leq \bar{\mu}_{\tilde{M}_1}(x(t))$, where

$$\underline{\mu}_{\tilde{M}_1}(x(t)) = \begin{cases} (1 - \sin(0.1x(t))) / 2, & \text{for } x(t) < 0 \\ (1 - \sin(0.5x(t))) / 2, & \text{for } x(t) \geq 0 \end{cases} \quad (10)$$

$$\bar{\mu}_{\tilde{M}_1}(x(t)) = \begin{cases} (1 - \sin(0.5x(t))) / 2, & \text{for } x(t) < 0 \\ (1 - \sin(0.1x(t))) / 2, & \text{for } x(t) \geq 0. \end{cases} \quad (11)$$

Similarly, we have $\underline{\mu}_{\tilde{M}_2}(x(t)) \leq \mu_{M_2^1}(x(t), a(x(t))) \leq \bar{\mu}_{\tilde{M}_2}(x(t))$, where

$$\underline{\mu}_{\tilde{M}_2}(x(t)) = \begin{cases} (1 + \sin(0.5x(t))) / 2, & \text{for } x(t) < 0 \\ (1 + \sin(0.1x(t))) / 2, & \text{for } x(t) \geq 0 \end{cases} \quad (12)$$

$$\bar{\mu}_{\tilde{M}_2}(x(t)) = \begin{cases} (1 + \sin(0.1x(t))) / 2, & \text{for } x(t) < 0 \\ (1 + \sin(0.5x(t))) / 2, & \text{for } x(t) \geq 0. \end{cases} \quad (13)$$

The following rules for the interval type-2 T-S fuzzy model can be achieved:

$$\text{Rule 1: If } x(t) \text{ is } \tilde{M}_1^i, \text{ then } \dot{x}(t) = A_i x(t). \quad (14)$$

From (5), the interval type-2 T-S fuzzy model for the nonlinear plant of (7) is defined as

$$\dot{x}(t) = \sum_{i=1}^2 (w_i^L(x(t)) \underline{v}_i(x(t)) + w_i^U(x(t)) \bar{v}_i(x(t))) A_i x(t) \quad (15)$$

where $\underline{\mu}_{\tilde{M}_i}(x(t)) = w_i^L(x(t))$, and $\bar{\mu}_{\tilde{M}_i}(x(t)) = w_i^U(x(t))$ for all i . Fig. 1(b) shows the type-1 membership function $\mu_{M_1^1}(x(t))$ with $a(t) = 0.35$, the lower and upper membership functions. The area in between the lower and upper membership functions is the FOU. Based on the lower and upper membership functions, the type-1 membership function in between can be reconstructed with the introduction of $\underline{v}_i(x(t))$ and $\bar{v}_i(x(t))$. Referring to Fig. 1(b), for $x(t) = -1.5$, it can be seen that the grades of membership at points A, B, and C are 0.8408, 0.7506, and 0.5747, respectively. The grade of membership at point B can be represented as $0.7506 = 0.5747(0.3390) + 0.8408(0.6610)$. By the same line of logic, every point of the membership function with $a(t) = 0.35$ can be determined based on the lower and upper membership functions. In general, the in-between membership functions can be reconstructed and represented in the form of a linear combination of the lower and upper membership functions, i.e., $w_i^L(x(t)) \underline{v}_i(x(t)) + w_i^U(x(t)) \bar{v}_i(x(t))$, where $\underline{v}_i(x(t)) + \bar{v}_i(x(t)) = 1$ for any $0.1 \leq a(x(t)) \leq 0.5$. In the above example, it can be seen that $\underline{v}_i(x(t)) = 0.3390$ and $\bar{v}_i(x(t)) = 0.6610$. In the stability analysis, we do know the exact values of $\underline{v}_i(x(t))$ and $\bar{v}_i(x(t))$, as they depend on the parameter uncertainty of $a(t)$. However, it can be seen that $\underline{v}_i(x(t))$ and $\bar{v}_i(x(t))$ exist.

With the interval type-2 fuzzy sets, the parameter uncertainties can be captured by the lower and upper membership functions. Consequently, an interval type-2 T-S fuzzy model of (15) can be used to describe the system of (7) with the uncertain parameter of $a(x(t))$. The interval type-2 T-S fuzzy model of (15) can be regarded as a collection of type-1 T-S fuzzy models with $a(x(t))$ taking various constant values.

Remark 1: The lower and upper membership functions for an interval type-2 T-S fuzzy model are not unique. Referring to Fig. 1(b), the FOU can be obtained by the area bounded by the lower and upper membership functions of $\underline{\mu}_{\tilde{M}_1^1}(x(t))$ and $\bar{\mu}_{\tilde{M}_1^1}(x(t))$, respectively. With the nonlinear functions of $\underline{v}_1(x(t))$ and $\bar{v}_1(x(t))$, the in-between membership functions can be reproduced. Based on this concept, by considering any arbitrary lower and upper membership functions denoted by $\underline{\eta}_{\tilde{M}_1^1}(x(t))$ and $\bar{\eta}_{\tilde{M}_1^1}(x(t))$ satisfying the conditions of $\underline{\eta}_{\tilde{M}_1^1}(x(t)) \leq \underline{\mu}_{\tilde{M}_1^1}(x(t))$ and $\bar{\eta}_{\tilde{M}_1^1}(x(t)) \geq \bar{\mu}_{\tilde{M}_1^1}(x(t))$, the actual membership function can be reconstructed with the nonlinear functions of $\underline{v}_1(x(t))$ and $\bar{v}_1(x(t))$ in other forms.

B. Interval Type-2 Fuzzy Controller

An interval type-2 fuzzy controller is proposed to stabilize the nonlinear plant based on the interval type-2 T-S fuzzy model of (5). The rule of the interval type-2 fuzzy controller is of the following format:

Rule j : If $f_1(\mathbf{x}(t))$ is $\tilde{M}_1^j$ and, $\dots$, and $f_\Psi(\mathbf{x}(t))$ is $\tilde{M}_\Psi^j$
then $\mathbf{u}(t) = \mathbf{G}_j \mathbf{x}(t)$ (16)

where $\mathbf{G}_j \in \mathbb{R}^{m \times n}$, $j = 1, 2, \dots, p$, are the feedback gains to be determined. The inferred output of the interval type-2 fuzzy controller is defined as

$$\mathbf{u}(t) = \sum_{j=1}^p (\underline{w}_j(\mathbf{x}(t)) + \bar{w}_j(\mathbf{x}(t))) \mathbf{G}_j \mathbf{x}(t) \quad (17)$$

where

$$\underline{w}_j(\mathbf{x}(t)) = \frac{w_j^L(\mathbf{x}(t))}{\sum_{k=1}^p (w_k^L(\mathbf{x}(t)) + w_k^U(\mathbf{x}(t)))} \geq 0$$

$$\bar{w}_j(\mathbf{x}(t)) = \frac{w_j^U(\mathbf{x}(t))}{\sum_{k=1}^p (w_k^L(\mathbf{x}(t)) + w_k^U(\mathbf{x}(t)))} \geq 0$$

which exhibits the property that $\sum_{j=1}^p (\underline{w}_j(\mathbf{x}(t)) + \bar{w}_j(\mathbf{x}(t))) = 1$. It can be seen from (17) that the type reduction for the proposed interval type-2 fuzzy controller is characterized by the average normalized membership grades of the lower and upper membership functions. Hence, in the following, the interval type-2 fuzzy controller of (17) is also referred to as the normalized central (NC) fuzzy controller.

III. STABILITY ANALYSIS OF INTERVAL TYPE-2 FUZZY-MODEL-BASED CONTROL SYSTEMS

In this section, the system stability of interval type-2 FMB control systems is investigated. LMI-based stability conditions

are derived using the Lyapunov-based approach. In the following analysis, $w_i(\mathbf{x}(t))$, $\underline{w}_j(\mathbf{x}(t))$, and $\bar{w}_j(\mathbf{x}(t))$ are denoted as w_i , $\underline{w}_j$, and $\bar{w}_j$, respectively, for brevity. Based on the property of the membership functions, we have $\sum_{i=1}^p w_i = \sum_{j=1}^p (\underline{w}_j + \bar{w}_j) = \sum_{i=1}^p \sum_{j=1}^p w_i (\underline{w}_j + \bar{w}_j) = 1$ to facilitate the stability analysis. To proceed to the stability analysis, we consider the interval type-2 FMB control system formed by the interval type-2 T-S fuzzy model of (5) and the type-2 fuzzy controller of (17) connected in a closed loop, which is shown as follows:

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \sum_{i=1}^p w_i \left(\mathbf{A}_i + \mathbf{B}_i \sum_{j=1}^p (\underline{w}_j + \bar{w}_j) \mathbf{G}_j \right) \mathbf{x}(t) \\ &= \sum_{i=1}^p \sum_{j=1}^p w_i (\underline{w}_j + \bar{w}_j) (\mathbf{A}_i + \mathbf{B}_i \mathbf{G}_j) \mathbf{x}(t). \end{aligned} \quad (18)$$

To investigate the system stability of (18), the following Lyapunov function candidate is considered:

$$V(t) = \mathbf{x}(t)^T \mathbf{P} \mathbf{x}(t) \quad (19)$$

where $\mathbf{P} = \mathbf{P}^T \in \mathbb{R}^{n \times n} > 0$. From (18) and (19), we have

$$\begin{aligned} \dot{V}(t) &= \dot{\mathbf{x}}(t)^T \mathbf{P} \mathbf{x}(t) + \mathbf{x}(t)^T \mathbf{P} \dot{\mathbf{x}}(t) \\ &= \left(\sum_{i=1}^p \sum_{j=1}^p w_i (\underline{w}_j + \bar{w}_j) (\mathbf{A}_i + \mathbf{B}_i \mathbf{G}_j) \mathbf{x}(t) \right)^T \mathbf{P} \mathbf{x}(t) \\ &\quad + \mathbf{x}(t)^T \mathbf{P} \sum_{i=1}^p \sum_{j=1}^p w_i (\underline{w}_j + \bar{w}_j) (\mathbf{A}_i + \mathbf{B}_i \mathbf{G}_j) \mathbf{x}(t) \\ &= \sum_{i=1}^p \sum_{j=1}^p w_i (\underline{w}_j + \bar{w}_j) \mathbf{x}(t) \\ &\quad \times ((\mathbf{A}_i + \mathbf{B}_i \mathbf{G}_j)^T \mathbf{P} + \mathbf{P} (\mathbf{A}_i + \mathbf{B}_i \mathbf{G}_j)) \mathbf{x}(t). \end{aligned} \quad (20)$$

Denote $\mathbf{X} = \mathbf{P}^{-1}$, $\mathbf{z}(t) = \mathbf{X}^{-1} \mathbf{x}(t)$, and $\mathbf{G}_j = \mathbf{N}_j \mathbf{X}^{-1}$, where $\mathbf{N}_j \in \mathbb{R}^{m \times n}$ for all j . From (20), we have

$$\begin{aligned} \dot{V}(t) &= \sum_{i=1}^p \sum_{j=1}^p w_i (\underline{w}_j + \bar{w}_j) \mathbf{z}(t)^T \mathbf{Q}_{ij} \mathbf{z}(t) \\ &= \mathbf{z}(t)^T \boldsymbol{\Theta} \mathbf{z}(t) \end{aligned} \quad (21)$$

where $\boldsymbol{\Theta} = \sum_{i=1}^p \sum_{j=1}^p w_i (\underline{w}_j + \bar{w}_j) \mathbf{Q}_{ij}$, and $\mathbf{Q}_{ij} = \mathbf{A}_i \mathbf{X} + \mathbf{X} \mathbf{A}_i^T + \mathbf{B}_i \mathbf{N}_j + \mathbf{N}_j^T \mathbf{B}_i^T$. It should be noted that (21) can represent a general form of fuzzy control systems subject to various problem formulations [10], for example, fuzzy control systems with decay rate, H_∞ or output feedback design. Once it is in the form of (21), the stability analysis below applies. It can be seen from (21) that $\dot{V}(t) \leq 0$ [the equality holds for $\mathbf{z}(t) = 0$] implies the asymptotic stability of the interval type-2 FMB control system of (18). To realize $\dot{V}(t) \leq 0$, it is required that $\boldsymbol{\Theta} < 0$. In the following, stability conditions for $\boldsymbol{\Theta} < 0$ are to be derived. To proceed to the stability analysis, based

on the property of the membership functions that $\sum_{i=1}^p w_i = \sum_{j=1}^p (\underline{w}_j + \bar{w}_j) = 1$, which leads to $\sum_{i=1}^p (w_i - \underline{w}_i - \bar{w}_i) = 0$, we have

$$\Xi = \sum_{i=1}^p \sum_{j=1}^p (w_i - \underline{w}_i - \bar{w}_i) (w_j (\mathbf{M}_j + \mathbf{M}_j^T) + \underline{w}_j (\mathbf{V}_j + \mathbf{V}_j^T) + \bar{w}_j (\mathbf{A}_j + \mathbf{A}_j^T)) = \mathbf{0} \quad (22)$$

where $\mathbf{M}_j \in \mathbb{R}^{n \times n}$, $\mathbf{V}_j \in \mathbb{R}^{n \times n}$, and $\mathbf{A}_j \in \mathbb{R}^{n \times n}$ are the slack matrices to relax the conservativeness of the stability analysis result. By expanding the terms in the right-hand side of (22), we have

$$\begin{aligned} \Xi = & \sum_{i=1}^p \sum_{j=1}^p w_i w_j (\mathbf{M}_j + \mathbf{M}_j^T) \\ & + \sum_{i=1}^p \sum_{j=1}^p w_i \underline{w}_j (\mathbf{V}_j + \mathbf{V}_j^T - \mathbf{M}_i - \mathbf{M}_i^T) \\ & + \sum_{i=1}^p \sum_{j=1}^p w_i \bar{w}_j (\mathbf{A}_j + \mathbf{A}_j^T - \mathbf{M}_i - \mathbf{M}_i^T) \\ & - \sum_{i=1}^p \sum_{j=1}^p \underline{w}_i \underline{w}_j (\mathbf{V}_j + \mathbf{V}_j^T) \\ & - \sum_{i=1}^p \sum_{j=1}^p \underline{w}_i \bar{w}_j (\mathbf{A}_j + \mathbf{A}_j^T + \mathbf{V}_i + \mathbf{V}_i^T) \\ & - \sum_{i=1}^p \sum_{j=1}^p \bar{w}_i \bar{w}_j (\mathbf{A}_j + \mathbf{A}_j^T). \end{aligned} \quad (23)$$

To further introduce some slack matrices in the stability analysis, the membership functions are assumed to satisfy the following inequalities:

$$-w_i + \rho_{i1} \underline{w}_i + \sigma_{i1} \bar{w}_i + \gamma_{i1} \geq 0, \quad \text{for all } i, \quad \mathbf{x}(t), \text{ and system parameters} \quad (24)$$

$$w_i - \rho_{i2} \underline{w}_i - \sigma_{i2} \bar{w}_i + \gamma_{i2} \geq 0, \quad \text{for all } i, \quad \mathbf{x}(t), \text{ and system parameters} \quad (25)$$

where ρ_{i1} , σ_{i1} , γ_{i1} , ρ_{i2} , σ_{i2} , and γ_{i2} are scalars to be determined. Defining $\mathbf{R}_{ij} + \mathbf{R}_{ij}^T \geq 0$, where $\mathbf{R}_{ij} = \mathbf{R}_{ji}^T \in \mathbb{R}^{n \times n}$, from (24) and (25), we have

$$\begin{aligned} \Phi = & \sum_{i=1}^p \sum_{j=1}^p (-w_i + \rho_{i1} \underline{w}_i + \sigma_{i1} \bar{w}_i + \gamma_{i1}) \\ & \times (w_j - \rho_{j2} \underline{w}_j - \sigma_{j2} \bar{w}_j + \gamma_{j2}) (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \geq 0. \end{aligned} \quad (26)$$

By expanding the terms in the right-hand side of (26), we have

$$\begin{aligned} \Phi = & \sum_{i=1}^p \sum_{j=1}^p (-w_i w_j + \rho_{j2} w_i \underline{w}_j + \sigma_{j2} w_i \bar{w}_j - w_i \gamma_{j2}) \\ & \times (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \end{aligned}$$

$$\begin{aligned} & + \sum_{i=1}^p \sum_{j=1}^p (\rho_{i1} \underline{w}_i w_j - \rho_{i1} \rho_{j2} \underline{w}_i \underline{w}_j - \rho_{i1} \sigma_{j2} \underline{w}_i \bar{w}_j \\ & \quad + \rho_{i1} \bar{w}_i \gamma_{j2}) (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \\ & + \sum_{i=1}^p \sum_{j=1}^p (\sigma_{i1} \bar{w}_i w_j - \sigma_{i1} \rho_{j2} \bar{w}_i \underline{w}_j - \sigma_{i1} \sigma_{j2} \bar{w}_i \bar{w}_j \\ & \quad + \sigma_{i1} \bar{w}_i \gamma_{j2}) (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \\ & + \sum_{i=1}^p \sum_{j=1}^p (\gamma_{i1} w_j - \gamma_{i1} \rho_{j2} \underline{w}_j - \gamma_{i1} \sigma_{j2} \bar{w}_j + \gamma_{i1} \gamma_{j2}) \\ & \quad \times (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \\ = & - \sum_{i=1}^p \sum_{j=1}^p w_i w_j (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \\ & - \sum_{i=1}^p \sum_{j=1}^p \underline{w}_i \underline{w}_j \rho_{i1} \rho_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \\ & - \sum_{i=1}^p \sum_{j=1}^p \bar{w}_i \bar{w}_j \sigma_{i1} \sigma_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \\ & + \sum_{i=1}^p \sum_{j=1}^p w_i \underline{w}_j (\rho_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) + \rho_{j1} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T)) \\ & + \sum_{i=1}^p \sum_{j=1}^p w_i \bar{w}_j (\sigma_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) + \sigma_{j1} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T)) \\ & - \sum_{i=1}^p \sum_{j=1}^p \underline{w}_i \bar{w}_j (\rho_{i1} \sigma_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \\ & \quad + \sigma_{j1} \rho_{i2} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T)) \\ & - \sum_{i=1}^p \sum_{j=1}^p w_i w_j \left(\sum_{k=1}^p \gamma_{k2} (\mathbf{R}_{ik} + \mathbf{R}_{ik}^T) \right. \\ & \quad \left. - \sum_{k=1}^p \sum_{l=1}^p \gamma_{k1} \gamma_{l2} (\mathbf{R}_{kl} + \mathbf{R}_{kl}^T) \right) \\ & + \sum_{i=1}^p \sum_{j=1}^p \underline{w}_i w_j \sum_{k=1}^p \rho_{i1} \gamma_{k2} (\mathbf{R}_{ik} + \mathbf{R}_{ik}^T) \\ & + \sum_{i=1}^p \sum_{j=1}^p \bar{w}_i w_j \sum_{k=1}^p \gamma_{k2} \sigma_{i1} (\mathbf{R}_{ik} + \mathbf{R}_{ik}^T) \\ & + \sum_{i=1}^p \sum_{j=1}^p w_i w_j \sum_{k=1}^p \gamma_{k1} (\mathbf{R}_{kj} + \mathbf{R}_{kj}^T) \\ & - \sum_{i=1}^p \sum_{j=1}^p w_i \underline{w}_j \sum_{k=1}^p \rho_{j2} \gamma_{k1} (\mathbf{R}_{kj} + \mathbf{R}_{kj}^T) \\ & - \sum_{i=1}^p \sum_{j=1}^p w_i \bar{w}_j \sum_{k=1}^p \sigma_{j2} \gamma_{k1} (\mathbf{R}_{kj} + \mathbf{R}_{kj}^T) \end{aligned}$$

$$\begin{aligned}
&= - \sum_{i=1}^p \sum_{j=1}^p w_i w_j \left(\mathbf{R}_{ij} + \mathbf{R}_{ij}^T + \sum_{k=1}^p \gamma_{k2} (\mathbf{R}_{ik} + \mathbf{R}_{ik}^T) \right. \\
&\quad - \sum_{k=1}^p \gamma_{k1} (\mathbf{R}_{kj} + \mathbf{R}_{kj}^T) \\
&\quad \left. - \sum_{k=1}^p \sum_{l=1}^p \gamma_{k1} \gamma_{l2} (\mathbf{R}_{kl} + \mathbf{R}_{kl}^T) \right) \\
&\quad - \sum_{i=1}^p \sum_{j=1}^p \underline{w}_i \underline{w}_j \rho_{i1} \rho_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \\
&\quad - \sum_{i=1}^p \sum_{j=1}^p \bar{w}_i \bar{w}_j \sigma_{i1} \sigma_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \\
&\quad + \sum_{i=1}^p \sum_{j=1}^p w_i \underline{w}_j \left(\rho_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) + \rho_{j1} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T) \right. \\
&\quad \left. - \sum_{k=1}^p \rho_{j2} \gamma_{k1} (\mathbf{R}_{kj} + \mathbf{R}_{kj}^T) \right. \\
&\quad \left. + \sum_{k=1}^p \rho_{j1} \gamma_{k2} (\mathbf{R}_{jk} + \mathbf{R}_{jk}^T) \right) \\
&\quad + \sum_{i=1}^p \sum_{j=1}^p w_i \bar{w}_j \left(\sigma_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) + \sigma_{j1} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T) \right. \\
&\quad \left. - \sum_{k=1}^p \sigma_{j2} \gamma_{k1} (\mathbf{R}_{kj} + \mathbf{R}_{kj}^T) \right. \\
&\quad \left. + \sum_{k=1}^p \gamma_{k2} \sigma_{j1} (\mathbf{R}_{jk} + \mathbf{R}_{jk}^T) \right) \\
&\quad - \sum_{i=1}^p \sum_{j=1}^p \underline{w}_i \bar{w}_j \left(\rho_{i1} \sigma_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \right. \\
&\quad \left. + \sigma_{j1} \rho_{i2} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T) \right) \\
&\quad + \sigma_{j1} \rho_{i2} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T) \Big). \tag{27}
\end{aligned}$$

$$\begin{aligned}
&- \sum_{i=1}^p \sum_{j=1}^p \bar{w}_i \bar{w}_j \left(\sigma_{i1} \sigma_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) + \mathbf{\Lambda}_j + \mathbf{\Lambda}_j^T \right) \\
&\quad + \sum_{i=1}^p \sum_{j=1}^p w_i \underline{w}_j \left(\rho_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) + \rho_{j1} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T) \right. \\
&\quad \left. - \sum_{k=1}^p \rho_{j2} \gamma_{k1} (\mathbf{R}_{kj} + \mathbf{R}_{kj}^T) \right. \\
&\quad \left. + \sum_{k=1}^p \rho_{j1} \gamma_{k2} (\mathbf{R}_{jk} + \mathbf{R}_{jk}^T) \right) \\
&\quad + \mathbf{Q}_{ij} + \mathbf{V}_j + \mathbf{V}_j^T - \mathbf{M}_i - \mathbf{M}_i^T \Big) \\
&\quad + \sum_{i=1}^p \sum_{j=1}^p w_i \bar{w}_j \left(\sigma_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) + \sigma_{j1} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T) \right. \\
&\quad \left. - \sum_{k=1}^p \sigma_{j2} \gamma_{k1} (\mathbf{R}_{kj} + \mathbf{R}_{kj}^T) \right. \\
&\quad \left. + \sum_{k=1}^p \gamma_{k2} \sigma_{j1} (\mathbf{R}_{jk} + \mathbf{R}_{jk}^T) \right) \\
&\quad + \mathbf{Q}_{ij} + \mathbf{\Lambda}_j + \mathbf{\Lambda}_j^T - \mathbf{M}_i - \mathbf{M}_i^T \Big) \\
&\quad - \sum_{i=1}^p \sum_{j=1}^p \underline{w}_i \bar{w}_j \left(\rho_{i1} \sigma_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \right. \\
&\quad \left. + \sigma_{j1} \rho_{i2} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T) \right. \\
&\quad \left. + \mathbf{\Lambda}_j + \mathbf{\Lambda}_j^T + \mathbf{V}_i + \mathbf{V}_i^T \right) \Big). \tag{28}
\end{aligned}$$

Define $\mathbf{S}_{ij} = \mathbf{S}_{ji}^T \in \mathbb{R}^{n \times n}$, $\mathbf{T}_{ij} \in \mathbb{R}^{n \times n}$, $\mathbf{U}_{ij} \in \mathbb{R}^{n \times n}$, $\mathbf{W}_{ij} \in \mathbb{R}^{n \times n}$, $\mathbf{Y}_{ij} = \mathbf{Y}_{ji}^T \in \mathbb{R}^{n \times n}$, and $\mathbf{Z}_{ij} = \mathbf{Z}_{ji}^T \in \mathbb{R}^{n \times n}$. Let

From (21), (23), and (27), we have

$$\begin{aligned}
\Theta &\leq \Theta + \Xi + \Phi \\
&= - \sum_{i=1}^p \sum_{j=1}^p w_i w_j \left(\mathbf{R}_{ij} + \mathbf{R}_{ij}^T + \sum_{k=1}^p \gamma_{k2} (\mathbf{R}_{ik} + \mathbf{R}_{ik}^T) \right. \\
&\quad - \sum_{k=1}^p \gamma_{k1} (\mathbf{R}_{kj} + \mathbf{R}_{kj}^T) \\
&\quad - \sum_{k=1}^p \sum_{l=1}^p \gamma_{k1} \gamma_{l2} \\
&\quad \left. \times (\mathbf{R}_{kl} + \mathbf{R}_{kl}^T) - \mathbf{M}_j - \mathbf{M}_j^T \right) \\
&\quad - \sum_{i=1}^p \sum_{j=1}^p \underline{w}_i \underline{w}_j \left(\rho_{i1} \rho_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) + \mathbf{V}_j + \mathbf{V}_j^T \right)
\end{aligned}$$

$$\begin{aligned}
\mathbf{S}_{ii} &< \mathbf{R}_{ii} + \mathbf{R}_{ii}^T + \sum_{k=1}^p \gamma_{k2} (\mathbf{R}_{ik} + \mathbf{R}_{ik}^T) \\
&\quad - \sum_{k=1}^p \gamma_{k1} (\mathbf{R}_{ki} + \mathbf{R}_{ki}^T) - \sum_{k=1}^p \sum_{l=1}^p \gamma_{k1} \gamma_{l2} \\
&\quad \times (\mathbf{R}_{kl} + \mathbf{R}_{kl}^T) - \mathbf{M}_i - \mathbf{M}_i^T, \quad i = 1, 2, \dots, p \tag{29}
\end{aligned}$$

$$\begin{aligned}
\mathbf{S}_{ij} + \mathbf{S}_{ij}^T &\leq \mathbf{R}_{ij} + \mathbf{R}_{ij}^T + \sum_{k=1}^p \gamma_{k2} (\mathbf{R}_{ik} + \mathbf{R}_{ik}^T) \\
&\quad - \sum_{k=1}^p \gamma_{k1} (\mathbf{R}_{kj} + \mathbf{R}_{kj}^T) - 2 \sum_{k=1}^p \sum_{l=1}^p \gamma_{k1} \gamma_{l2} (\mathbf{R}_{kl} + \mathbf{R}_{kl}^T) \\
&\quad - \mathbf{M}_j - \mathbf{M}_j^T - \mathbf{M}_i - \mathbf{M}_i^T + \mathbf{R}_{ji} \\
&\quad + \mathbf{R}_{ji}^T + \sum_{k=1}^p \gamma_{k2} (\mathbf{R}_{jk} + \mathbf{R}_{jk}^T) \\
&\quad - \sum_{k=1}^p \gamma_{k1} (\mathbf{R}_{ki} + \mathbf{R}_{ki}^T), \quad j = 1, 2, \dots, p; i < j \tag{30}
\end{aligned}$$

$$\begin{aligned} \mathbf{T}_{ij} + \mathbf{T}_{ij}^T &\geq \rho_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \\ &+ \rho_{j1} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T) - \sum_{k=1}^p \rho_{j2} \gamma_{k1} (\mathbf{R}_{kj} + \mathbf{R}_{kj}^T) \\ &+ \sum_{k=1}^p \rho_{j1} \gamma_{k2} (\mathbf{R}_{jk} + \mathbf{R}_{jk}^T) + \mathbf{Q}_{ij} + \mathbf{V}_j + \mathbf{V}_j^T \\ &- \mathbf{M}_i - \mathbf{M}_i^T, \quad i, j = 1, 2, \dots, p \end{aligned} \quad (31)$$

$$\begin{aligned} \mathbf{U}_{ij} + \mathbf{U}_{ij}^T &\geq \sigma_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) + \sigma_{j1} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T) \\ &- \sum_{k=1}^p \sigma_{j2} \gamma_{k1} (\mathbf{R}_{kj} + \mathbf{R}_{kj}^T) + \sum_{k=1}^p \gamma_{k2} \sigma_{j1} (\mathbf{R}_{jk} + \mathbf{R}_{jk}^T) \\ &+ \mathbf{Q}_{ij} + \mathbf{\Lambda}_j + \mathbf{\Lambda}_j^T - \mathbf{M}_i - \mathbf{M}_i^T, \quad i, j = 1, 2, \dots, p \end{aligned} \quad (32)$$

$$\begin{aligned} \mathbf{W}_{ij} + \mathbf{W}_{ij}^T &\leq \rho_{i1} \sigma_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) + \sigma_{j1} \rho_{i2} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T) \\ &+ \mathbf{\Lambda}_j + \mathbf{\Lambda}_j^T + \mathbf{V}_i + \mathbf{V}_i^T, \quad i, j = 1, 2, \dots, p \end{aligned} \quad (33)$$

$$\mathbf{Y}_{ii} < \rho_{i1} \rho_{i2} (\mathbf{R}_{ii} + \mathbf{R}_{ii}^T) + \mathbf{V}_i + \mathbf{V}_i^T, \quad i = 1, 2, \dots, p \quad (34)$$

$$\begin{aligned} \mathbf{Y}_{ij} + \mathbf{Y}_{ij}^T &\leq \rho_{i1} \rho_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) + \mathbf{V}_j + \mathbf{V}_j^T \\ &+ \rho_{j1} \rho_{i2} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T) + \mathbf{V}_i + \mathbf{V}_i^T, \quad j = 1, 2, \dots, p; \quad i < j \end{aligned} \quad (35)$$

$$\mathbf{Z}_{ii} < \sigma_{i1} \sigma_{i2} (\mathbf{R}_{ii} + \mathbf{R}_{ii}^T) + \mathbf{\Lambda}_i + \mathbf{\Lambda}_i^T, \quad i = 1, 2, \dots, p \quad (36)$$

$$\begin{aligned} \mathbf{Z}_{ij} + \mathbf{Z}_{ij}^T &\leq \sigma_{i1} \sigma_{j2} (\mathbf{R}_{ij} + \mathbf{R}_{ij}^T) \\ &+ \mathbf{\Lambda}_j + \mathbf{\Lambda}_j^T + \sigma_{j1} \sigma_{i2} (\mathbf{R}_{ji} + \mathbf{R}_{ji}^T) \\ &+ \mathbf{\Lambda}_i + \mathbf{\Lambda}_i^T, \quad j = 1, 2, \dots, p; \quad i < j. \end{aligned} \quad (37)$$

From (28)–(37), we have

$$\begin{aligned} \dot{V}(t) &\leq - \sum_{i=1}^p w_i^2 \mathbf{z}(t)^T \mathbf{S}_{ii} \mathbf{z}(t) \\ &- \sum_{j=1}^p \sum_{i < j} w_i w_j \mathbf{z}(t)^T (\mathbf{S}_{ij} + \mathbf{S}_{ij}^T) \mathbf{z}(t) \\ &- \sum_{i=1}^p \underline{w}_i^2 \mathbf{z}(t)^T \mathbf{Y}_{ii} \mathbf{z}(t) \\ &- \sum_{j=1}^p \sum_{i < j} \underline{w}_i \underline{w}_j \mathbf{z}(t)^T (\mathbf{Y}_{ij} + \mathbf{Y}_{ij}^T) \mathbf{z}(t) \\ &- \sum_{i=1}^p \bar{w}_i^2 \mathbf{z}(t)^T \mathbf{Z}_{ii} \mathbf{z}(t) \\ &- \sum_{j=1}^p \sum_{i < j} \bar{w}_i \bar{w}_j \mathbf{z}(t)^T (\mathbf{Z}_{ij} + \mathbf{Z}_{ij}^T) \mathbf{z}(t) \\ &+ \sum_{i=1}^p \sum_{j=1}^p w_i w_j \mathbf{z}(t)^T (\mathbf{T}_{ij} + \mathbf{T}_{ij}^T) \mathbf{z}(t) \\ &+ \sum_{i=1}^p \sum_{j=1}^p w_i \bar{w}_j \mathbf{z}(t)^T (\mathbf{U}_{ij} + \mathbf{U}_{ij}^T) \mathbf{z}(t) \\ &- \sum_{i=1}^p \sum_{j=1}^p \underline{w}_i \bar{w}_j \mathbf{z}(t)^T (\mathbf{W}_{ij} + \mathbf{W}_{ij}^T) \mathbf{z}(t) \\ &= \begin{bmatrix} \mathbf{r}(t) \\ \mathbf{s}(t) \\ \mathbf{t}(t) \end{bmatrix}^T \begin{bmatrix} -\mathbf{S} & \mathbf{T} & \mathbf{U} \\ \mathbf{T}^T & -\mathbf{Y} & -\mathbf{W} \\ \mathbf{U}^T & -\mathbf{W}^T & -\mathbf{Z} \end{bmatrix} \begin{bmatrix} \mathbf{r}(t) \\ \mathbf{s}(t) \\ \mathbf{t}(t) \end{bmatrix} \end{aligned} \quad (38)$$

where

$$\begin{aligned} \mathbf{r}(t) &= \begin{bmatrix} w_1 \mathbf{z}(t) \\ w_2 \mathbf{z}(t) \\ \vdots \\ w_p \mathbf{z}(t) \end{bmatrix} \\ \mathbf{s}(t) &= \begin{bmatrix} \underline{w}_1 \mathbf{z}(t) \\ \underline{w}_2 \mathbf{z}(t) \\ \vdots \\ \underline{w}_p \mathbf{z}(t) \end{bmatrix} \\ \mathbf{t}(t) &= \begin{bmatrix} \bar{w}_1 \mathbf{z}(t) \\ \bar{w}_2 \mathbf{z}(t) \\ \vdots \\ \bar{w}_p \mathbf{z}(t) \end{bmatrix} \\ \mathbf{S} &= \begin{bmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} & \cdots & \mathbf{S}_{1p} \\ \mathbf{S}_{21} & \mathbf{S}_{22} & \cdots & \mathbf{S}_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{S}_{p1} & \mathbf{S}_{p2} & \cdots & \mathbf{S}_{pp} \end{bmatrix} \\ \mathbf{T} &= \begin{bmatrix} \mathbf{T}_{11} & \mathbf{T}_{12} & \cdots & \mathbf{T}_{1p} \\ \mathbf{T}_{21} & \mathbf{T}_{22} & \cdots & \mathbf{T}_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{T}_{p1} & \mathbf{T}_{p2} & \cdots & \mathbf{T}_{pp} \end{bmatrix} \\ \mathbf{U} &= \begin{bmatrix} \mathbf{U}_{11} & \mathbf{U}_{12} & \cdots & \mathbf{U}_{1p} \\ \mathbf{U}_{21} & \mathbf{U}_{22} & \cdots & \mathbf{U}_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{U}_{p1} & \mathbf{U}_{p2} & \cdots & \mathbf{U}_{pp} \end{bmatrix} \\ \mathbf{W} &= \begin{bmatrix} \mathbf{W}_{11} & \mathbf{W}_{12} & \cdots & \mathbf{W}_{1p} \\ \mathbf{W}_{21} & \mathbf{W}_{22} & \cdots & \mathbf{W}_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{W}_{p1} & \mathbf{W}_{p2} & \cdots & \mathbf{W}_{pp} \end{bmatrix} \\ \mathbf{Y} &= \begin{bmatrix} \mathbf{Y}_{11} & \mathbf{Y}_{12} & \cdots & \mathbf{Y}_{1p} \\ \mathbf{Y}_{21} & \mathbf{Y}_{22} & \cdots & \mathbf{Y}_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{Y}_{p1} & \mathbf{Y}_{p2} & \cdots & \mathbf{Y}_{pp} \end{bmatrix} \\ \mathbf{Z} &= \begin{bmatrix} \mathbf{Z}_{11} & \mathbf{Z}_{12} & \cdots & \mathbf{Z}_{1p} \\ \mathbf{Z}_{21} & \mathbf{Z}_{22} & \cdots & \mathbf{Z}_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{Z}_{p1} & \mathbf{Z}_{p2} & \cdots & \mathbf{Z}_{pp} \end{bmatrix}. \end{aligned}$$

From (38), it can be seen that for

$$\begin{bmatrix} -\mathbf{S} & \mathbf{T} & \mathbf{U} \\ \mathbf{T}^T & -\mathbf{Y} & -\mathbf{W} \\ \mathbf{U}^T & -\mathbf{W}^T & -\mathbf{Z} \end{bmatrix} < 0$$

we have $\dot{V}(t) \leq 0$ [the equality holds when $\mathbf{z}(t) = \mathbf{0}$], which implies the asymptotic stability of the interval type-2 FMB control system of (18). The stability analysis result is summarized in the following theorem.

Theorem 1: The interval type-2 FMB control system of (18), formed by the interval type-2 fuzzy model of (5) and the interval type-2 fuzzy controller of (17) connected in a closed loop, is guaranteed to be asymptotically stable if there exist predefined scalars of ρ_{i1} , σ_{i1} , γ_{i1} , ρ_{i2} , σ_{i2} , and γ_{i2} such that the inequalities of $-w_i(\mathbf{x}(t)) + \rho_{i1}\underline{w}_i(\mathbf{x}(t)) + \sigma_{i1}\bar{w}_i(\mathbf{x}(t)) + \gamma_{i1} \geq 0$ and $w_i(\mathbf{x}(t)) - \rho_{i2}\underline{w}_i(\mathbf{x}(t)) - \sigma_{i2}\bar{w}_i(\mathbf{x}(t)) + \gamma_{i2} \geq 0$ are satisfied for all i , $\mathbf{x}(t)$, and system parameters, and there exist matrices of $\mathbf{M}_j \in \mathbb{R}^{n \times n}$, $\mathbf{N}_j \in \mathbb{R}^{m \times n}$, $\mathbf{R}_{ij} = \mathbf{R}_{ji}^T \in \mathbb{R}^{n \times n}$, $\mathbf{S}_{ij} = \mathbf{S}_{ji}^T \in \mathbb{R}^{n \times n}$, $\mathbf{T}_{ij} \in \mathbb{R}^{n \times n}$, $\mathbf{U}_{ij} \in \mathbb{R}^{n \times n}$, $\mathbf{V}_j \in \mathbb{R}^{n \times n}$, $\mathbf{W}_{ij} \in \mathbb{R}^{n \times n}$, $\mathbf{X} = \mathbf{X}^T \in \mathbb{R}^{n \times n}$, $\mathbf{Y}_{ij} = \mathbf{Y}_{ji}^T \in \mathbb{R}^{n \times n}$, $\mathbf{Z}_{ij} = \mathbf{Z}_{ji}^T \in \mathbb{R}^{n \times n}$, and $\mathbf{\Lambda}_j \in \mathbb{R}^{n \times n}$ such that the following LMIs are satisfied: $\mathbf{R}_{ij} + \mathbf{R}_{ij}^T \geq 0$, $i, j = 1, 2, \dots, p$; (29)–(37); and

$$\begin{bmatrix} -\mathbf{S} & \mathbf{T} & \mathbf{U} \\ \mathbf{T}^T & -\mathbf{Y} & -\mathbf{W} \\ \mathbf{U}^T & -\mathbf{W}^T & -\mathbf{Z} \end{bmatrix} < 0.$$

The feedback gains are defined as $\mathbf{G}_j = \mathbf{N}_j \mathbf{X}^{-1}$, $j = 1, 2, \dots, p$.

IV. SIMULATION EXAMPLES

In this section, two simulation examples are presented to illustrate the effectiveness of the proposed approach.

A. Numerical Example

Consider a fuzzy model with three fuzzy rules in the following format:

Rule i : If $x_1(t)$ is M_1^i ,
then $\dot{\mathbf{x}}(t) = \mathbf{A}_i \mathbf{x}(t) + \mathbf{B}_i u(t)$, $i = 1, 2, 3$ (39)

where

$$\begin{aligned} \mathbf{A}_1 &= \begin{bmatrix} 2.78 & -5.63 \\ 0.01 & 0.33 \end{bmatrix} \\ \mathbf{A}_2 &= \begin{bmatrix} 0.2 & -3.22 \\ 0.35 & 0.12 \end{bmatrix} \\ \mathbf{A}_3 &= \begin{bmatrix} -a & -6.63 \\ 0.45 & 0.15 \end{bmatrix} \\ \mathbf{B}_1 &= \begin{bmatrix} 2 \\ -1 \end{bmatrix} \\ \mathbf{B}_2 &= \begin{bmatrix} 8 \\ 0 \end{bmatrix} \\ \mathbf{B}_3 &= \begin{bmatrix} -b + 6 \\ -1 \end{bmatrix} \end{aligned}$$

where $22 \leq a \leq 30$ and $20 \leq b \leq 25$. The fuzzy model is assumed working in the operating domain of $x_1(t) \in [-10 \ 10]^T$. For illustration purposes, the lower membership functions are chosen as $w_1^L(x_1(t)) = 0.95 - (0.925/(1 + e^{-((x+3.5)/8)}))$, $w_3^L(x_1(t)) = 0.025 + (0.925/(1 + e^{-((x-3.5)/8)}))$, and $w_2^L(x_1(t)) = 1 - w_1^L(x_1(t)) - w_3^L(x_1(t))$.

The upper membership functions are chosen as $w_1^U(x_1(t)) = 0.95 - (0.925/(1 + e^{-((x+4.5)/8)}))$, $w_3^U(x_1(t)) = 0.025 + (0.925/(1 + e^{-((x-4.5)/8)}))$, and $w_2^U(x_1(t)) = 1 - w_1^U(x_1(t)) - w_3^U(x_1(t))$. The actual membership functions subject to parameter uncertainties are inside the regions bounded by the lower and upper membership functions.

In this example, with the NC fuzzy controller defined in (17), it is assumed that $\rho_{i1} = \sigma_{i1} = 2$, $\rho_{i2} = \sigma_{i2} = 0.01$, $\gamma_{i1} = -0.025$, and $\gamma_{i2} = -0.024$ such that the inequalities of (24) and (25) are satisfied. By employing the stability conditions in Theorem 1, the stability region is shown in Fig. 2(a) indicated by open circles. To show the influence of the slack variables $\mathbf{M}_i$, $\mathbf{V}_i$, and $\mathbf{\Lambda}_i$ to the stability conditions, the stability region without these slack variables is shown in Fig. 2(a) indicated by crosses. It can be seen from Fig. 2(a) that the stability conditions can be relaxed with these slack variables. As the membership functions are uncertain, the type-1 stability conditions in [5]–[10] cannot be applied. For comparison purposes, the stability conditions in [4] and [19], which do not require the exact membership function information, are employed to check for the stability regions. The stability region indicated by crosses for [4] is shown in Fig. 1(b). The stability region indicated by open circles for [19] is shown in Fig. 1(b), in which the parameters required by the stability conditions are chosen as $\rho_{j1} = 0.62$ and $\rho_{j2} = 2.5$. Referring to Fig. 1, it can be seen that the proposed stability conditions under the interval type-2 fuzzy framework outperform some published type-1 ones in terms of a larger stability region. To investigate the influence of the bounding coefficients for the inequalities of (24) and (25), we choose γ_{i1} as -0.048 and -0.012 with other coefficients unchanged to check for the stability region for the proposed stability conditions, which is shown in Fig. 2(c). It can be seen that different bounding coefficients (due to different lower and upper membership functions) will lead to different sizes of stability regions.

B. Inverted Pendulum

An inverted pendulum [18] subject to parameter uncertainty is considered as the nonlinear plant. In this paper, for illustration purposes, the inverted pendulum without considering the cart movement is considered. Based on the interval type-2 T-S fuzzy model representing the inverted pendulum, the proposed NC fuzzy controller is employed to stabilize the inverted pendulum. The dynamics of the inverted pendulum are described by the following dynamical equations:

$$\ddot{\theta}(t) = \frac{g \sin(\theta(t)) - a m_p L \dot{\theta}(t)^2 \sin(2\theta(t)) / 2 - a \cos(\theta(t)) u(t)}{4L/3 - a m_p L \cos^2(\theta(t))} \quad (40)$$

where $\theta(t)$ is the angular displacement of the pendulum, $g = 9.8 \text{ m/s}^2$ is the acceleration due to gravity, $m_p \in [m_{p_{\min}} \ m_{p_{\max}}] = [2 \ 3] \text{ kg}$ is the mass of the pendulum, $M_c \in [M_{c_{\min}} \ M_{c_{\max}}] = [8 \ 16] \text{ kg}$ is the mass of the cart, $a = 1/(m_p + M_c)$, $2L = 1 \text{ m}$ is the length of the pendulum, and $u(t)$ is the force applied to the cart. m_p and M_c are regarded as the parameter uncertainties. It was reported in [17] that the

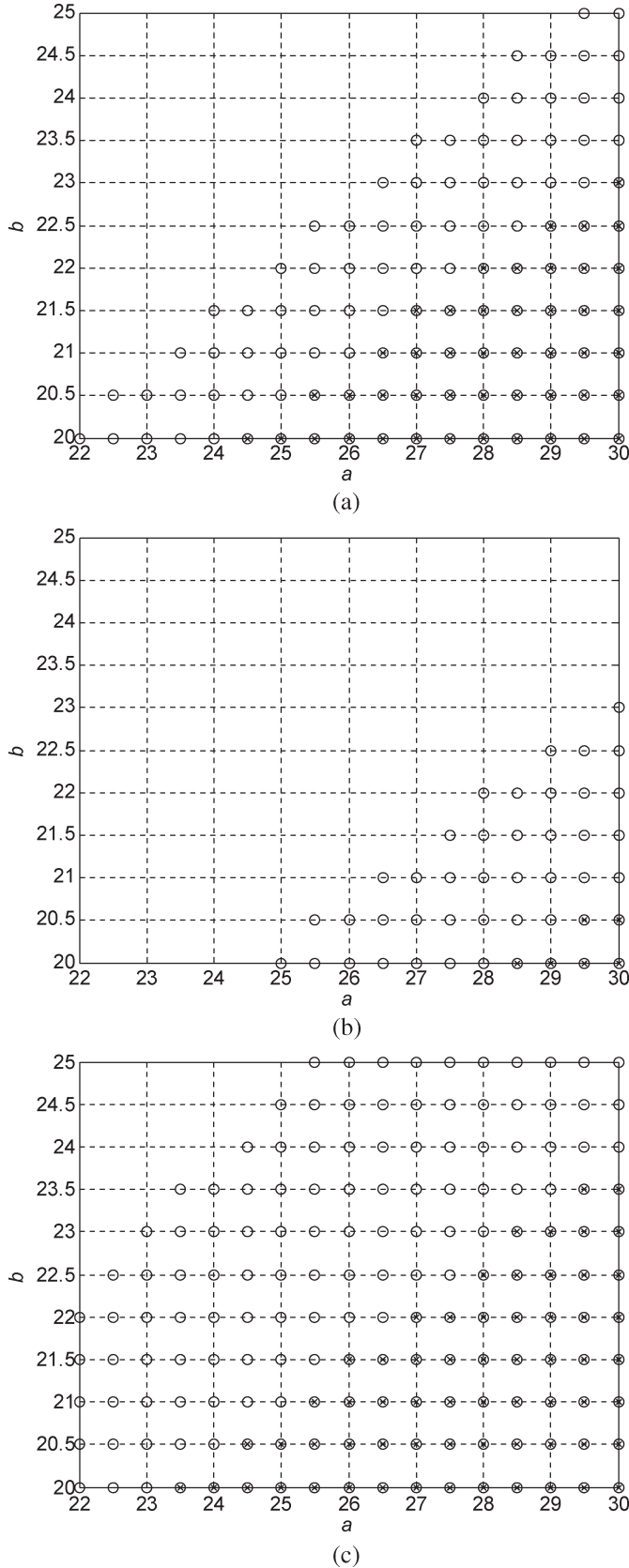


Fig. 2. Stability regions for the proposed and published stability conditions. (a) Stability regions for the proposed stability conditions with (open circles) and without (crosses) M_i , V_i , and A_i . (b) Stability regions for [19] (open circles) and [4] (crosses). (c) Stability regions of the proposed stability conditions with different bounding coefficients with $\gamma_{i1} = -0.048$ (open circles) and $\gamma_{i1} = -0.012$ (crosses).

uncertainty-free inverted pendulum, of which m_p and M_c take constant values, can be exactly represented by the following type-1 T-S fuzzy model:

$$\dot{\mathbf{x}}(t) = \sum_{i=1}^4 w_i(\mathbf{x}(t)) (\mathbf{A}_i \mathbf{x}(t) + \mathbf{B}_i u(t)) \quad (41)$$

where

$$\mathbf{x}(t) = [x_1(t) \quad x_2(t)]^T = [\theta(t) \quad \dot{\theta}(t)]^T$$

$$x_1(t) \in [x_{1\min} \quad x_{1\max}] = [-\frac{5\pi}{12} \quad \frac{5\pi}{12}]$$

$$x_2(t) \in [x_{2\min} \quad x_{2\max}] = [-5 \quad 5]$$

$$\mathbf{A}_1 = \mathbf{A}_2 = \begin{bmatrix} 0 & 1 \\ f_{1\min} & 0 \end{bmatrix}$$

$$\mathbf{A}_3 = \mathbf{A}_4 = \begin{bmatrix} 0 & 1 \\ f_{1\max} & 0 \end{bmatrix}$$

$$\mathbf{B}_1 = \mathbf{B}_3 = \begin{bmatrix} 0 \\ f_{2\min} \end{bmatrix}$$

$$\mathbf{B}_2 = \mathbf{B}_4 = \begin{bmatrix} 0 \\ f_{2\max} \end{bmatrix}$$

$$f_{1\min} = 10.0078$$

$$f_{1\max} = 18.4800$$

$$f_{2\min} = -0.0261 \times 10^{-3}$$

$$f_{2\max} = -0.1765$$

$$w_i(\mathbf{x}(t)) = \frac{\mu_{M_1^i}(f_1(\mathbf{x}(t))) \times \mu_{M_2^i}(f_2(x_1(t))))}{\sum_{l=1}^4 (\mu_{M_1^l}(f_1(\mathbf{x}(t))) \times \mu_{M_2^l}(f_2(x_1(t))))}$$

$$\mu_{M_1^1}(f_1(\mathbf{x}(t))) = \mu_{M_1^2}(f_1(\mathbf{x}(t))) = \frac{-f_1(\mathbf{x}(t)) + f_{1\max}}{f_{1\max} - f_{1\min}}$$

$$\mu_{M_1^3}(f_1(\mathbf{x}(t))) = \mu_{M_1^4}(f_1(\mathbf{x}(t))) = 1 - \mu_{M_1^1}(f_1(\mathbf{x}(t)))$$

$$\mu_{M_2^1}(f_2(x_1(t))) = \mu_{M_2^2}(f_2(x_1(t))) = \frac{-f_2(x_1(t)) + f_{2\max}}{f_{2\max} - f_{2\min}}$$

$$\mu_{M_2^3}(f_2(x_1(t))) = \mu_{M_2^4}(f_2(x_1(t))) = 1 - \mu_{M_2^1}(f_2(x_1(t)))$$

$$f_1(\mathbf{x}(t)) = \frac{g - am_p L x_2(t)^2 \cos(x_1(t))}{4L/3 - am_p L \cos^2(x_1(t))} \left(\frac{\sin(x_1(t))}{x_1(t)} \right)$$

$$f_2(x_1(t)) = \frac{-a \cos(x_1(t))}{4L/3 - am_p L \cos^2(x_1(t))}.$$

It can be seen from (41) that the grade of membership becomes uncertain in value when m_p and M_c are uncertain. Consequently, the stability conditions in [5]–[10] cannot be applied to design a stable fuzzy controller. In this example, an interval type-2 T-S fuzzy model with four rules is employed to describe the inverted pendulum subject to parameter uncertainties. Referring to (1), the rule is of the following format:

Rule i : If $x_1(t)$ is $\tilde{M}_1^i$ and $x_2(t)$ is $\tilde{M}_2^i$, then

$$\dot{\mathbf{x}}(t) = \mathbf{A}_i \mathbf{x}(t) + \mathbf{B}_i u(t), \quad i = 1, 2, 3, 4. \quad (42)$$

TABLE I
LOWER AND UPPER MEMBERSHIP FUNCTIONS FOR THE INTERVAL TYPE-2 T-S FUZZY MODEL OF THE INVERTED PENDULUM

Lower and upper membership functions	
$\underline{\mu}_{\tilde{M}_1^1}(x_1(t)) = \underline{\mu}_{\tilde{M}_1^2}(x_1(t)) =$ $\frac{-f_1(\mathbf{x}(t)) + f_{1_{\max}}}{f_{1_{\max}} - f_{1_{\min}}}; \underline{\mu}_{\tilde{M}_1^1}(x_1(t)) =$ $\underline{\mu}_{\tilde{M}_1^4}(x_1(t)) = \frac{f_1(\mathbf{x}(t)) - f_{1_{\min}}}{f_{1_{\max}} - f_{1_{\min}}} \text{ with } x_2(t) =$ $0, m_p = m_{p_{\max}} \text{ and } M_c = M_{c_{\min}}$	$\bar{\mu}_{\tilde{M}_2^1}(f_2(x_1(t))) = \bar{\mu}_{\tilde{M}_2^3}(f_2(x_1(t))) =$ $\frac{-f_2(x_1(t)) + f_{2_{\max}}}{f_{2_{\max}} - f_{2_{\min}}}; \bar{\mu}_{\tilde{M}_2^1}(f_2(x_1(t))) =$ $\bar{\mu}_{\tilde{M}_2^4}(f_2(x_1(t))) = \frac{f_2(x_1(t)) - f_{2_{\min}}}{f_{2_{\max}} - f_{2_{\min}}} \text{ with}$ $m_p = m_{p_{\max}} \text{ and } M_c = M_{c_{\max}}$
$\underline{\mu}_{\tilde{M}_1^3}(x_1(t)) = \underline{\mu}_{\tilde{M}_1^4}(x_1(t)) =$ $\frac{f_1(\mathbf{x}(t)) - f_{1_{\min}}}{f_{1_{\max}} - f_{1_{\min}}}; \underline{\mu}_{\tilde{M}_1^3}(x_1(t)) =$ $\underline{\mu}_{\tilde{M}_1^2}(x_1(t)) = \frac{-f_1(\mathbf{x}(t)) + f_{1_{\max}}}{f_{1_{\max}} - f_{1_{\min}}} \text{ with } x_2(t)$ $= x_{2_{\max}}, m_p = m_{p_{\max}} \text{ and } M_c = M_{c_{\min}}$	$\bar{\mu}_{\tilde{M}_2^2}(f_2(x_1(t))) = \bar{\mu}_{\tilde{M}_2^4}(f_2(x_1(t))) =$ $\frac{f_2(x_1(t)) - f_{2_{\min}}}{f_{2_{\max}} - f_{2_{\min}}}; \bar{\mu}_{\tilde{M}_2^2}(f_2(x_1(t))) =$ $\bar{\mu}_{\tilde{M}_2^3}(f_2(x_1(t))) = \frac{-f_2(x_1(t)) + f_{2_{\max}}}{f_{2_{\max}} - f_{2_{\min}}}$ $\text{with } m_p = m_{p_{\min}} \text{ and } M_c = M_{c_{\min}}$

Referring to (5), the interval type-2 T-S fuzzy model is defined as follows:

$$\dot{\mathbf{x}}(t) = \sum_{i=1}^4 (w_i^L(x_1(t)) \underline{v}_i(x_1(t)) + w_i^U(x_1(t)) \bar{v}_i(x_1(t))) \times (\mathbf{A}_i \mathbf{x}(t) + \mathbf{B}_i u(t)). \quad (43)$$

When m_p and M_c are uncertain but bounded, the interval type-2 T-S fuzzy model of (43) is regarded as a collection of a number of type-1 T-S fuzzy models in the form of (41), covering the whole ranges of m_p and M_c .

To determine the lower and upper membership functions, it is necessary to satisfy the following conditions:

$$\begin{aligned} \underline{\mu}_{\tilde{M}_1^i}(x_1(t)) &\leq \mu_{M_1^i}(f_1(\mathbf{x}(t))) \\ &\leq \bar{\mu}_{\tilde{M}_1^i}(x_1(t)), \quad i = 1, 2, 3, 4 \end{aligned} \quad (44)$$

$$\begin{aligned} \underline{\mu}_{\tilde{M}_2^i}(x_1(t)) &\leq \mu_{M_2^i}(f_2(x_1(t))) \\ &\leq \bar{\mu}_{\tilde{M}_2^i}(x_1(t)), \quad i = 1, 2, 3, 4. \end{aligned} \quad (45)$$

By taking $x_2(t)$, m_c , and M_c as constant values in their operating ranges of $f_1(\mathbf{x}(t))$ and $f_2(x_1(t))$, and considering all their combinations, we can determine the lower and upper bounds of $\mu_{M_1^i}(f_1(\mathbf{x}(t)))$ and $\mu_{M_2^i}(f_2(x_1(t)))$, which are taken as the lower and upper membership functions of the interval type-2 T-S fuzzy model. In this example, the lower and upper membership functions are defined in Table I. Fig. 3 shows the plots of the lower and upper membership functions. The lower

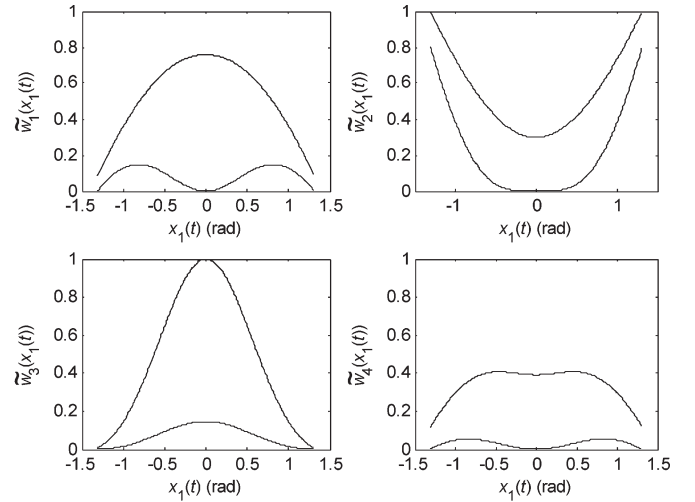


Fig. 3. Lower and upper membership functions of $\tilde{w}_i(x_1(t))$, $i = 1, 2, 3, 4$.

and upper grades of membership for each rule are respectively defined as

$$\begin{aligned} w_i^L(x_1(t)) &= \frac{\underline{\mu}_{\tilde{M}_1^i}(x_1(t)) \times \underline{\mu}_{\tilde{M}_2^i}(x_1(t))}{\sum_{l=1}^4 (\underline{\mu}_{\tilde{M}_1^l}(x_1(t)) \times \underline{\mu}_{\tilde{M}_2^l}(x_1(t)))} \\ w_i^U(x_1(t)) &= \frac{\bar{\mu}_{\tilde{M}_1^i}(x_1(t)) \times \bar{\mu}_{\tilde{M}_2^i}(x_1(t))}{\sum_{l=1}^4 (\bar{\mu}_{\tilde{M}_1^l}(x_1(t)) \times \bar{\mu}_{\tilde{M}_2^l}(x_1(t)))} \end{aligned}$$

for all i . It can be shown that $w_i^L(x_1(t)) \leq w_i(x_1(t)) \leq w_i^U(x_1(t))$. It is worth noting that any functions in the range

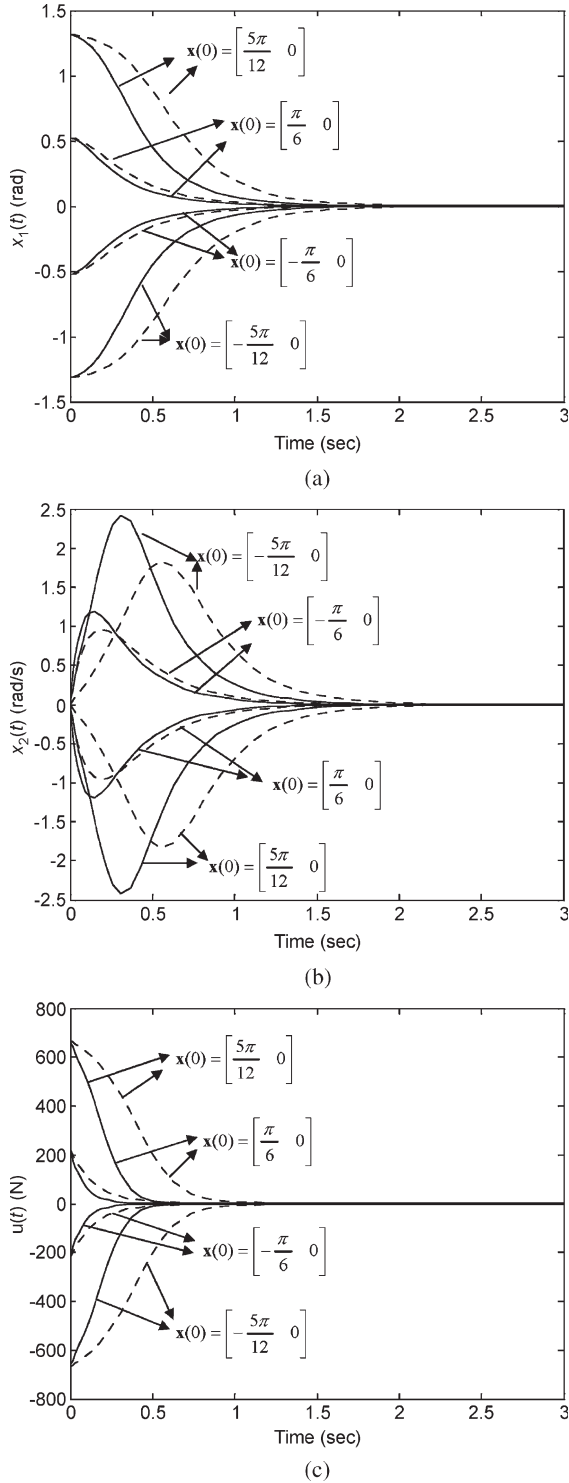


Fig. 4. System-state responses and control signals for the inverted pendulum. (solid lines) $m_p = m_{p_{\min}}$ and $M_c = M_{c_{\min}}$. (dotted line) $m_p = m_{p_{\max}}$ and $M_c = M_{c_{\max}}$. (a) $x_1(t)$. (b) $x_2(t)$. (c) $u(t)$.

of 0 to 1 satisfying the conditions of (44) and (45) can be taken as the lower and upper membership functions. As a result, the interval type-2 T-S fuzzy model for a nonlinear system is not unique.

Based on the interval type-2 T-S fuzzy model, a four-rule NC fuzzy controller is employed to stabilize the inverted pendulum

of (40), i.e., to drive $\mathbf{x}(t)$ to $\mathbf{0}$ as time goes by. The rule of the NC fuzzy controller is of the following format:

Rule j : If $x_1(t)$ is $\tilde{M}_1^j$ and $x_1(t)$ is $\tilde{M}_2^j$,
then $u(t) = \mathbf{G}_j \mathbf{x}(t)$, $j = 1, 2, 3, 4$. (46)

Referring to (17), the NC fuzzy controller is defined as follows:

$$u(t) = \frac{\sum_{j=1}^4 (w_j^L(x_1(t)) + w_j^U(x_1(t))) \mathbf{G}_j \mathbf{x}(t)}{\sum_{k=1}^4 (w_k^L(x_1(t)) + w_k^U(x_1(t)))}. \quad (47)$$

It can be shown that the membership function conditions of (24) and (25) hold with $\rho_{ik} = 1$, $\sigma_{ik} = 0.001$, $\gamma_{11} = 0.441$, $\gamma_{21} = 0.191$, $\gamma_{31} = 0.525$, $\gamma_{41} = 0.2$, and $\gamma_{i2} = 0.001$, for $i = 1, 2, 3, 4$, $k = 1, 2$. Based on Theorem 1, with the MATLAB LMI toolbox, we have $\mathbf{G}_1 = [934.7822 \ 238.7590]$, $\mathbf{G}_2 = [1027.3843 \ 260.9691]$, $\mathbf{G}_3 = [1000.4873 \ 254.0293]$, and $\mathbf{G}_4 = [1138.3668 \ 289.6253]$. The type-2 fuzzy controller is employed to control the inverted pendulum of (40). Fig. 4 shows the system-state responses and control signals of the interval type-2 FMB control system with $m_p = m_{p_{\min}}$ and $M_c = M_{c_{\min}}$ under the initial system states of $\mathbf{x}(0) = [\frac{5\pi}{12} \ 0]$, $[-\frac{5\pi}{12} \ 0]$, $[\frac{\pi}{6} \ 0]$, and $[-\frac{\pi}{6} \ 0]$. To check for the robustness of the NC fuzzy controller, we set $m_p = m_{p_{\max}}$ and $M_c = M_{c_{\max}}$. It can be seen from Fig. 4 that the NC fuzzy controller is able to stabilize the inverted pendulum with different parameter values of m_p and M_c .

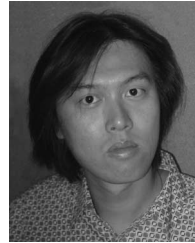
V. CONCLUSION

An interval type-2 T-S fuzzy model has been proposed to represent the nonlinear plant. With the interval type-2 fuzzy sets characterized by the lower and upper membership functions, the parameter uncertainties of the nonlinear plant can be effectively captured. An interval type-2 fuzzy controller has been proposed to close the feedback loop to form an interval type-2 FMB control system. The system stability has been investigated using the Lyapunov-based approach. Based on the information of the lower and upper membership functions of the interval type-2 T-S fuzzy model, some free matrices are introduced to facilitate the stability analysis. LMI-based stability conditions have been derived to guarantee the system stability and aid the design of the interval type-2 fuzzy controller. Simulation examples have been given to illustrate the effectiveness of the proposed approach.

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